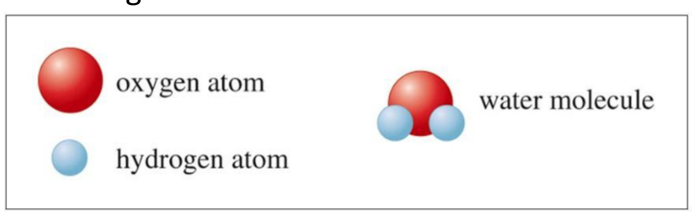
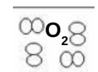
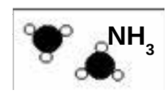
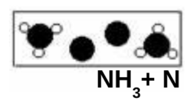
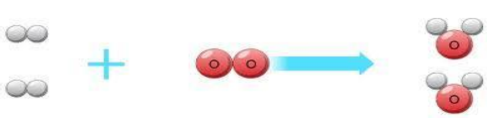
Name______________________________________HR______________________

8th Grade Science Unit 4 Changing Matter STAR Cards!

QUESTION	ANSWER
4a) 1. What is a physical change? 2. Name 2 examples.	4a) 1. A physical change **changes the look** of a substance; but **does NOT create a new substance** 2. Examples ● Broken glass ● Boiling water ● Chopping wood
4b) 1. What is a chemical change? 2. Name 1 example	4b) 1. A chemical change **creates a new substance** through a chemical reaction 2. Examples ● Iron turning to rust ● Burning wood into ashes ● Mixing baking soda and vinegar
4c) 1. What is an atom? 2. What is a molecule?	4c) 1. An atom is the smallest building block of matter. It is the smallest part of an element that is still that element. 2. A molecule is a bunch of atoms bonded together



QUESTION	ANSWER
4d) Water is a **compound** and has the chemical symbol: (H_2O) In chemical symbols of compounds like water (H_2O)… ● What do the <u>letters</u> tell you? ● What do the <u>numbers</u> tell you?	4d) ● The <u>letters</u> tell you the **elements the compound** is made of. *(H=Hydrogen, O=Oxygen)* ● The <u>numbers</u> tell you **how many atoms of each element** there are. *(ex. H_2 has 2 atoms of Hydrogen)*
4e) How can you tell if something is an: <u>element</u>, <u>compound</u> or <u>mixture</u>?	4e) ● An <u>element</u> has one type of atom. (1 capital letter in the formula)  ● A <u>compound</u> has more than one type of atom. (more than one capital letter in the formula)  ● A <u>mixture</u> can be separated. (Plus sign (+) in the formula) 
4f) 1) What happens to <u>particles</u> when they are <u>heated</u>? 2) What happens to <u>particles</u> when they are <u>cooled</u>?	4f) 1) Particles **speed up** and **spread out** when they are heated. 2) Particles **slow down** and **come closer together** when they are cooled.
4g) 1) What are the <u>reactants</u> in a chemical reaction? 2) What are <u>products</u> in a chemical reaction? 	4g) 1) The <u>**reactants**</u> are the ingredients or **what goes into a chemical reaction.** 2) The <u>**products**</u> are what **comes out (**what is made) from a chemical reaction. 

QUESTION	ANSWER
4h) 1)What does the Law of Conservation of Mass say? 2) What does this mean when thinking about chemical reactions?	4h) 1)The Law of Conservation of Mass says that **matter cannot be created or destroyed,** (it can only change form). 2)The **products** will have the **same mass and number and type of atoms** as the reactants.